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MASTER'S THESIS | Trabajo Fin de Máster

Title of Your Master's Thesis, Exactly as in the Approved Proposal

An optional subtitle that adds context

[Title of the thesis in the second language]

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[City], June 2026

Title of Your Master's Thesis, Exactly as in the Approved Proposal: An optional subtitle that adds context

Master's Thesis, Master's Degree in [Name of the Master's Programme], Universidad de [University Name], 2025/2026.

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Supervisor's Approval

Dr. First-name Surname and Dr. First-name Surname, supervisors of the Master's Thesis entitled *Title of Your Master's Thesis, Exactly as in the Approved Proposal*, carried out by **First-name Surname1 Surname2** within the Master's Degree in [Name of the Master's Programme], hereby authorise its submission for public defence and assessment at the [School or Faculty Name], Universidad de [University Name].

[City], June 2026

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Declaration of Originality

I, **First-name Surname1 Surname2**, student of the Master's Degree in [Name of the Master's Programme] at Universidad de [University Name], declare that:

1. This Master's Thesis, entitled *Title of Your Master's Thesis, Exactly as in the Approved Proposal*, is my own original work and has been carried out under the supervision of Dr. First-name Surname and Dr. First-name Surname.
2. It has not been submitted, in whole or in part, for any other degree or qualification at this or any other university.
3. All the sources used have been properly cited and acknowledged, and every figure, table or text taken from other works has been reproduced with permission or under the terms of its licence.
4. Any use of generative artificial intelligence tools has been declared in this document in accordance with the regulations of my university.
5. I am aware that plagiarism and any other form of academic dishonesty may lead to the failure of this Master's Thesis and to disciplinary measures.

[City], June 2026

First-name Surname1 Surname2

Author

To [whoever you want to dedicate this work to].

Acknowledgements

Template guidance

Thank the people and institutions that made this work possible: your supervisors, colleagues, family and friends. If the work was funded (a grant, a research project, a company internship) or used shared resources (a computing cluster, a dataset, a laboratory), acknowledge them here with the exact name and reference number that the funder requires. Keep it to one page.

I would like to thank my supervisors, Dr. First-name Surname and Dr. First-name Surname, for their guidance and support throughout this project. [Continue with your own acknowledgements.]

Abstract

Template guidance

The abstract is the most widely read part of your thesis, and often the only one that appears in your university's repository. Write it last, as a self-contained text of about 200–300 words (check the limit of your university) that covers, in this order: the *context* and the problem, the *objective*, the *methods*, the main *results* (with numbers when possible) and the *conclusions*. Avoid citations, undefined acronyms, equations and references to figures or tables.

This document is a LaTeX template for Master's theses written in English at Spanish public universities, and at the same time it is its own user guide. It provides a complete and configurable structure, from the cover page and the declarations commonly required by Spanish regulations to the chapters of a research-oriented thesis, the references and the appendices. All the personal and institutional data are defined in a single configuration file, so the same template can be used at any university by changing the university name, logos, colours and a few options. The template supports digital and printed versions, several typefaces, a second abstract in Spanish or in any co-official language, a review mode with line numbers for supervisors, and guidance boxes that disappear automatically in the final version. Replace this text with your own abstract.

Keywords: Master's thesis; LaTeX; template; Spanish universities; reproducibility

Resumen

[Title of the thesis in the second language]

> **Template guidance**
>
> Most Spanish universities require the title, the abstract and the keywords also in Spanish (or in the co-official language of the region) when the thesis is written in English. Write here a faithful translation of your English abstract. The text inside this environment is hyphenated according to the rules of that language. The title and the keywords come from `\ThesisTitleSecondLanguage` and `\KeywordsSecondLanguage` in `config.tex`.

[Write here the translation of your abstract into the second language.]

Palabras clave: [keyword 1; keyword 2; keyword 3]

Statement on the Use of Generative AI

Template guidance

Be specific: name each tool and its version, explain what you used it for and in which parts of the thesis, and state that you have reviewed the output and take full responsibility for the content. Some universities only allow AI for language editing, others require your supervisor's authorisation for other uses. If you did not use any generative AI tool, keep only the alternative sentence below.

In accordance with the regulations of Universidad de [University Name], I declare that I have used the generative artificial intelligence tools listed in Table 1, for the purposes and in the parts of this Master's Thesis indicated there.

Table 1. Generative AI tools used in the preparation of this thesis.

Tool (version)	Purpose	Parts affected
[Tool name] ([version])	Language editing of drafts written by the author	Chapters 1–3
[Tool name] ([version])	Help with debugging the analysis code	Code repository

I have critically reviewed and, where necessary, edited all the output of these tools. The research questions, methodology, analysis, interpretation of the results and conclusions are my own, and I take full responsibility for the content of this work.

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List of Acronyms

DOI Digital Object Identifier

ECTS European Credit Transfer and Accumulation System

GDPR General Data Protection Regulation

GPU graphics processing unit

ML machine learning

SDG Sustainable Development Goal

TFM Trabajo Fin de Máster

CHAPTER 1

Using This Template

This chapter explains how to use this LaTeX template to write a Master's thesis (*Trabajo Fin de Máster (TFM)*) in English at a Spanish university. The template is designed for `pdflatex`, the most widely used LaTeX compiler, together with `biber` for the bibliography, and it also works with `lualatex`.

You can compile the sources with a local LaTeX distribution, such as TeX Live (`https://tug.org/texlive/`) or MiKTeX (`https://miktex.org/`), or online with Overleaf (`https://www.overleaf.com`), which needs no installation. The template is an English, Master's-level and university-independent adaptation of the TFG template created by Francisco Charte Ojeda for the Escuela Politécnica Superior de Jaén (`https://github.com/fcharte/TFGLatexTemplate`).

Template guidance

Boxes like this one contain guidance about what to write in each part of the thesis. They are only visible while you write: add the `final` option to the `\documentclass` line of `main.tex` and all of them disappear from the PDF, together with the to-do notes. When you start writing your own thesis, hide this whole chapter by setting `tfmTemplateGuide` to `false` in `config.tex`.

1.1 Before you start: your university's regulations

In Spain, Master's degrees are regulated by Royal Decree 822/2021 [1], which establishes that every Master's degree ends with a thesis worth between 6 and 30 European Credit Transfer and Accumulation System (ECTS) credits that must be defended in public. Beyond that common framework, each university, and often each school or Master's programme, sets its own rules: length, structure, formatting, the language in which the thesis can be written, whether an abstract in Spanish or in a co-official language is required, the declarations that must be signed, and how the thesis is submitted and archived.

This template implements the requirements that are most common across Spanish public universities and lets you adapt the rest. Before writing, read the regulations of your university and your Master's programme, and check in particular:

- whether you need *permission* to write the thesis in English, and whether the title, abstract, keywords or conclusions must also be written in Spanish or in a co-official language (option `secondlanguage`);
- whether there is an *official cover* that must be used (field `\OfficialCover`) and which logos may appear;
- the *formatting* rules: margins, typeface, size and line spacing;
- the *declarations* required: originality, supervisor's approval, use of generative artificial intelligence, alignment with the Sustainable Development Goals (SDGs);
- the *maximum length* and the citation style expected in your field.

1.2 Getting started

1.2.1 Files of the template

Table 1.1 describes the files and folders of the template. In most cases you only need to edit `config.tex`, the files in `frontmatter/` and `chapters/`, and `references.bib`.

Table 1.1. Files and folders of the template.

File or folder	Purpose
`main.tex`	Main document: class options and order of the parts of the thesis.
`config.tex`	Your data: title, author, supervisors, university, logos, licence, optional pages.
`tfmthesis.cls`	The class that defines the layout. You do not need to edit it.
`frontmatter/`	Dedication, acknowledgements, abstracts, AI statement and acronyms.
`chapters/`	One file per chapter.
`appendices/`	One file per appendix.
`figures/`	Images and plots (`figures/src/` holds the scripts that generate them, `figures/logos/` the logos).
`tables/`	Long tables kept in separate files.
`references.bib`	Bibliographic references in BibLaTeX format.
`presets/`	Settings that reproduce the style of specific universities.

1.2.2 Setting your data

Open `config.tex` and fill in your data, following the comments in the file. Listing 1.1 shows its most important fields. These data are used automatically on the cover, on the declarations, in the page headers and in the metadata of the PDF file.

```
1 \ThesisTitle{Title of Your Master's Thesis}
```

```
\ThesisShortTitle{Short title for the page headers}
\ThesisTitleSecondLanguage{Título del trabajo en español}
\ThesisAuthor{First-name Surname1 Surname2}
\Supervisor{Dr. First-name Surname}{Department of ..., University of ...}
\Supervisor[Co-supervisor]{Dr. First-name Surname}{Company or institute}
\University{Universidad de ...}
\Faculty{School of ...}
\Degree{Master's Degree in ...}
\AcademicYear{2025/2026}
\SubmissionDate{June 2026}
\UniversityLogo{figures/logos/university-logo.pdf}
\definecolor{tfmaccent}{HTML}{1F4E79}
```

Listing 1.1. Main fields of `config.tex`.

The `\Supervisor` command can be repeated as many times as needed, and its optional argument sets the role shown on the cover (Supervisor, Co-supervisor, External supervisor. . .). If a field is left empty, it disappears from the document. In your own text you can reuse these data with `\thesistitle`, `\thesisauthor`, `\thesisdegree`, `\thesisuniversity` and `\thesissupervisors`.

1.2.3 Class options

The appearance of the document is controlled with the options of the `\documentclass` command in `main.tex`, summarised in Table 1.2. For example, the final version of a thesis written in a sans-serif typeface, to be printed, with the abstract also in Catalan and references in APA style, would start with:

```
\documentclass[print, final, font=sans, secondlanguage=catalan,
               bibstyle=apa]{tfmthesis}
```

Listing 1.2. Example of class options.

1.2.4 Optional pages

Table 1.3 lists the optional pages of the front matter and how to include or remove them. The order in which they appear is set in `main.tex`.

1.2.5 Compiling the document

The bibliography is processed with `biber`, so a full compilation needs several steps. The simplest way is to let `latexmk` run them for you:

```
latexmk main.tex          # builds main.pdf (pdfLaTeX + Biber)
latexmk -pvc main.tex     # recompiles automatically whenever a file changes
```

Table 1.2. Options of the `tfmthesis` class. Default values are shown in bold.

Option	Effect
`print`	Two-sided layout for a printed and bound copy: wider inner margin, chapters start on odd pages and links are black. Without it the document is optimised for the screen (**digital**).
`final`	Hides the guidance boxes and the to-do notes.
`review`	Adds line numbers and a 'DRAFT' watermark, so that your supervisors can refer to specific lines in their comments.
`font=`	**`serif`** (Times-like), `sans` (Helvetica/Arial-like) or `modern` (Latin Modern, the classic LaTeX look).
`secondlanguage=`	Language of the second abstract: **`spanish`**, `catalan`, `valencian`, `galician`, `basque` or `none`.
`bibstyle=`	Any BibLaTeX style: **`numeric-comp`**, `ieee`, `apa`, `authoryear`, `nature`…
`english=`	**`british`** or `american` spelling and hyphenation.
`uppercase`	Chapter titles in capital letters.
`11pt`	Smaller body text (**12pt** by default).

Table 1.3. Optional pages of the front matter.

Page	How to control it
Official cover	`\OfficialCover{file.pdf}` in `config.tex`
Licence / copyright	Toggle `tfmLicencePage`; licence chosen with `\ThesisLicence`
Supervisor's approval	Toggle `tfmApprovalPage`
Declaration of originality	Toggle `tfmOriginalityPage`
Dedication, AI statement, abstracts	Comment out their `\input` line in `main.tex`
Lists of figures, tables, algorithms, listings and acronyms	Comment out the corresponding line in `main.tex`

```
latexmk -c                # removes the auxiliary files
```

Listing 1.3. Compiling the thesis from the command line.

Overleaf, TeXstudio, VS Code with LaTeX Workshop and most editors detect all of this automatically. In Overleaf, keep the compiler set to pdfLaTeX (Menu → Compiler).

1.3 Structuring the thesis

A thesis is similar to a book, so its content is organised in several levels with a hierarchical numbering, which you can see in the table of contents. The usual levels are:

1. Chapter (`\chapter{Title}`)
    1.1. Section (`\section{Title}`)
        1.1.1. Subsection (`\subsection{Title}`)
            1.1.1.1. Subsubsection (`\subsubsection{Title}`)

More than four levels are not recommended. Subsubsections are numbered but, to keep the table of contents short, they are not listed in it; add `\setcounter{tocdepth}{3}` to `config.tex` to include them.

1.3.1 Chapters

Each chapter is written in a separate `.tex` file inside `chapters/`, which is included in `main.tex` with `\input{chapters/file}`. The first line of each chapter defines its title and a label to refer to it:

```
\chapter{Materials and Methods}\label{ch:methods}
```

Listing 1.4. Beginning of a chapter.

Write the titles in title case; the numbering is automatic. If your university requires chapter titles in capitals, use the `uppercase` class option instead of typing them in capitals, so that the table of contents and the headers stay consistent.

1.3.2 Sections and subsections

Plan the sections of a chapter before writing it: think about what you want to tell and how to split it into cohesive parts, each of which becomes a section. Long sections are divided into

subsections. In general, several pages of text without any division make the document hard to follow.

1.3.3 Paragraphs

Paragraphs are separated by a blank line in the source. The text of a paragraph is written continuously, without pressing Enter: LaTeX breaks lines and hyphenates words automatically. Do not use `\\` or `\newline` to separate paragraphs. It is good practice to have at least one paragraph between two consecutive headings, so that a section never starts immediately after the title of its chapter.

1.3.4 Cross-references

Referring to other parts of the document (chapters, sections, figures, tables, equations. . .) takes two steps:

1. Put a label right after the element, with `\label{key}`. Using a prefix that identifies the type of element (`ch:`, `sec:`, `fig:`, `tab:`, `eq:`, `lst:`, `alg:`) makes keys easier to remember.
2. Refer to it with `\cref{key}`, which inserts both the type of element and its number, e.g. 'Section 1.3.1'. Use `\Cref` at the beginning of a sentence.

The `\nameref` command inserts the title of the element and `\cpageref` its page. For example[1]:

```
\Cref{sec:chapters}, \nameref{sec:chapters}, on \cpageref{sec:chapters},
explains how to split the thesis into chapters.
```

Listing 1.5. Cross-references.

produces: 'Section 1.3.1, Chapters, on Page 5, explains how to split the thesis into chapters.' All references are hyperlinks in the PDF and are updated automatically when the document changes.

1.4 Text formatting

The styles of headings and paragraphs are predefined. In the text you will mainly use three variants:

- **Bold**, with `\textbf{...}`: use it sparingly to **highlight something important**; if there is too much bold text, it loses its effect.

[1] The ~ character is a non-breaking space: it keeps the words on both sides on the same line.

- *Italics*, with `\textit{...}` or `\emph{...}`: for terms in other languages, e.g. *Trabajo Fin de Máster*, for titles of works and occasionally for emphasis.
- `Monospace`, with `\texttt{...}`: for names of variables, functions, files and anything related to code, e.g. the `print()` function of Python.

LaTeX allows colours, font sizes and many other effects, but a thesis is a formal document: use them only when they convey information. Quotation marks are written with `\enquote{...}`, which produces the correct 'curly quotes' for the language.

1.4.1 Footnotes

Footnotes[2] are useful to clarify something or add complementary information without breaking the flow of the text. They are inserted with `\footnote{Text}` right where the number must appear. Like bold text, do not abuse them.

1.4.2 Lists

Lists make enumerations and procedures with several steps easier to read. Use a numbered list (`enumerate`) when the order matters, and a bulleted list (`itemize`) otherwise. In LaTeX lists are *environments*: blocks of content that start with `\begin{name}` and end with `\end{name}`. Each element starts with `\item`, as shown in Listing 1.6. Lists can be nested, and the indentation, bullets and numbering adjust automatically.

```
\begin{itemize}
  \item \texttt{itemize}: produces a bulleted list
  \item \texttt{enumerate}: produces a numbered list
\end{itemize}
```

Listing 1.6. A bulleted list with two items.

1.5 Figures

A thesis contains many kinds of figures: diagrams of a system, plots of results, screenshots, photographs... Keep the following rules in mind:

- Every figure has a number and a *caption* that describes clearly what it shows. A long caption is not a problem.

[2] Footnotes are numbered automatically within each chapter.

- Every figure must be referred to from the text (‘as shown in Figure 1.1. . . ’). A figure that is not mentioned should not be there.
- **Every figure taken or adapted from another source must indicate its origin** and comply with its licence (see Section 1.11.2). Figures without a source are assumed to be your own.
- Figures must be legible: their text should be roughly the size of the caption, and they should never look blurry or pixelated.

1.5.1 Formats and files

Whenever possible, use *vector* formats (PDF, EPS, SVG converted to PDF) for plots and diagrams: they keep their quality at any zoom level. Use PNG for screenshots and JPEG for photographs, with enough resolution (at least 300 dpi at the printed size). Save your figures in `figures/` with descriptive names, and keep the scripts that generate them in `figures/src/`, so that they can be regenerated if the data change: the plots of this chapter are produced by `figures/src/example_figures.py`.

1.5.2 Inserting a figure

Figures are inserted with `\includegraphics` inside a `figure` environment, which also contains the `\caption` and the `\label`. Use widths relative to the text width, such as `width=0.8\textwidth`. Listing 1.7 produces Figure 1.1.

```latex
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.85\textwidth]{example-learning-curves}
  \caption{Training and validation loss of a model. The dashed line marks the
           epoch with the lowest validation loss.}
  \label{fig:learning-curves}
\end{figure}
```

Listing 1.7. Inserting a figure.

The optional argument `[htbp]` lets LaTeX place the figure *here*, at the *top* or *bottom* of a page, or on a separate *page*, wherever it fits best. Do not fight this behaviour: floats moving to the next page are normal, and you just refer to them by number.

1.5.3 Subfigures

Several related images can be grouped in a single figure with the `subfigure` environment, each with its own caption and label, as in Figure 1.2. You can refer to the whole figure or to each part, e.g. Figure 1.2a.

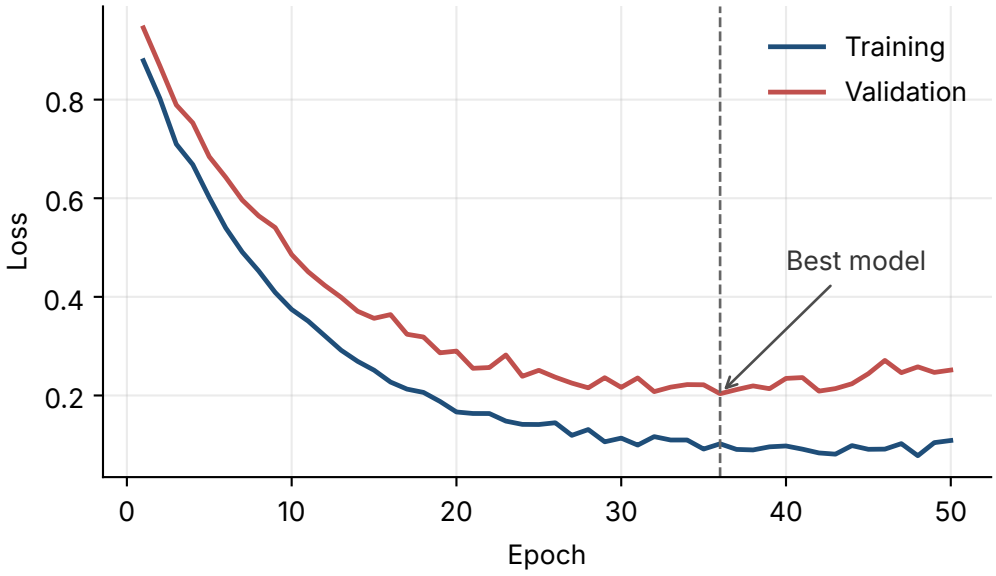
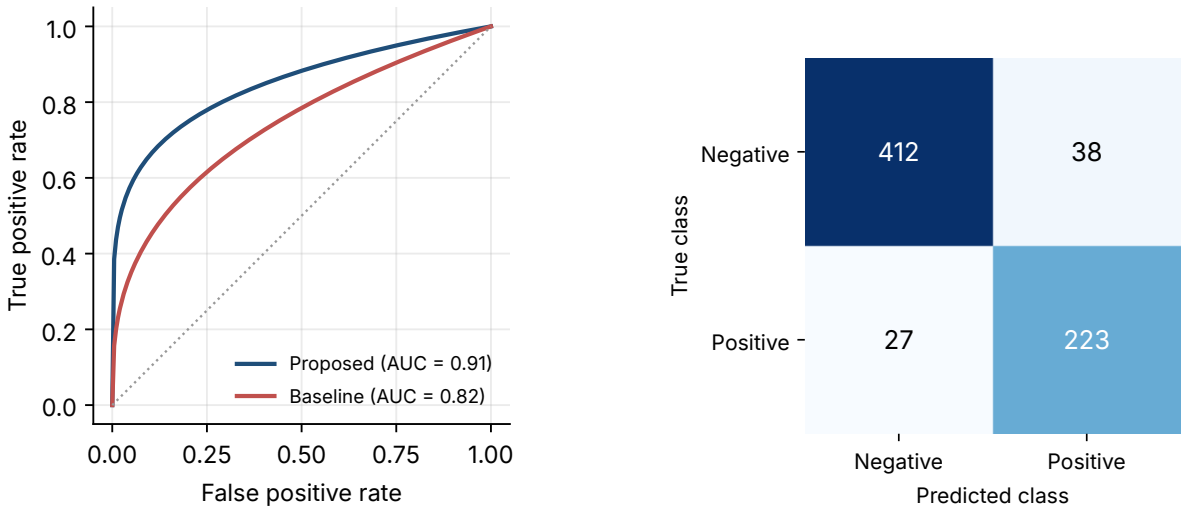


Figure 1.1. Training and validation loss of a model. The dashed line marks the epoch with the lowest validation loss.



(a) ROC curves.

(b) Confusion matrix of the proposed model.

Figure 1.2. Example of a figure with two subfigures.

1.5.4 Diagrams drawn with TikZ

Diagrams can also be drawn directly in LaTeX with the TikZ package, so that their fonts match the text exactly. Figure 1.3 is an example. Alternatively, draw diagrams with tools such as draw.io (`https://app.diagrams.net`) or PlantUML (`https://plantuml.com`) and export them as PDF.

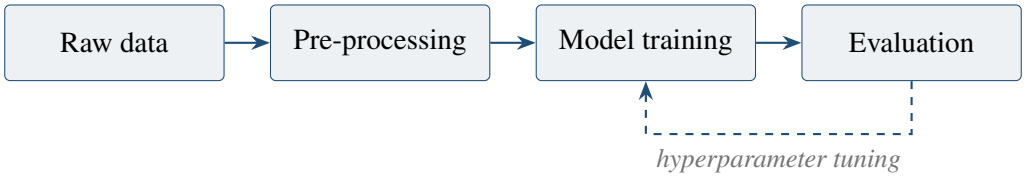


Figure 1.3. A typical machine learning pipeline drawn with TikZ.

1.6 Tables

Tables are very common in a thesis, especially to report experimental results. The rules for figures apply to tables too: every table has a number, a caption and a reference from the text. In addition:

- The caption of a table goes *above* it (the template does it automatically when `\caption` is written before the `tabular`).
- All tables share the same format, with three horizontal rules (`\toprule`, `\midrule` and `\bottomrule` from the `booktabs` package), column headings in bold and no vertical lines.
- Numbers are aligned on the decimal point and reported with a sensible number of decimal places.

Tables are created with the `tabular` environment inside a `table` float. The argument of `tabular` gives the alignment of each column: `l` (left), `c` (centre), `r` (right) or `p{5cm}` (fixed width with line wrapping). The `tabularx` environment adds the `X` column type, which takes all the remaining width, very convenient for columns with long text. Columns are separated with `&` and rows end with `\\`. Listing 1.8 produces Table 1.4.

```
\begin{table}[htbp]
  \centering
  \caption{Datasets used in the experiments.}
  \label{tab:example}
  \begin{tabular}{@{}l r r S[table-format=2.1]@{}}
    \toprule
    \textbf{Dataset} & \textbf{Instances} & \textbf{Features} & {\textbf{Missing (\%)}} \\
    \midrule
    Hospital A & 12,480 & 54  & 3.2  \\
```

```
    Hospital B &  8,915 & 54  & 11.7 \\
    Public     & 70,000 & 112 & 0.4  \\
    \bottomrule
  \end{tabular}
\end{table}
```

Listing 1.8. Creating a table.

Table 1.4. Datasets used in the experiments.

Dataset	Instances	Features	Missing (%)
Hospital A	12,480	54	3.2
Hospital B	8,915	54	11.7
Public	70,000	112	0.4

The `S` column type of the `siunitx` package aligns numbers on the decimal point, as in the last column of Table 1.4; headings in `S` columns must be enclosed in braces. The same package typesets units consistently: `\SI{8}{\giga\byte}` gives 8 GB and `\num{0.05}` gives 0.05. Table 6.1 in Chapter 6 shows a complete results table, and Table 5.1 shows a table stored in its own file (`tables/hardware-requirements.tex`) and included with `\input`, which is handy for long tables or tables generated by a program.

1.7 Equations

LaTeX excels at typesetting mathematics. There are two cases:

- *Inline* expressions, within a paragraph, are delimited by dollar signs: `$f(n) = n^2 + 1$` produces $f(n) = n^2 + 1$.
- *Displayed* equations, on their own line, numbered and labelled so that they can be referenced. They use the `equation` environment, or `align` for several aligned equations.

Listing 1.9 produces Equation (1.1), the binary cross-entropy loss.

```
\begin{equation}
  \mathcal{L}(\bm{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} \left[ y_i \log \hat{y}_i
                             + (1 - y_i) \log (1 - \hat{y}_i) \right]
  \label{eq:cross-entropy}
\end{equation}
```

Listing 1.9. A displayed equation.

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i) \right] \tag{1.1}$$

Equations are part of the text, so they are punctuated like any other sentence, and every symbol must be defined the first time it appears: here N is the number of examples, y_i the true label of the i-th example and $\hat{y}_i$ the predicted probability. The Wikibooks guide (https://en.wikibooks.org/wiki/LaTeX/Mathematics) answers most questions about mathematical notation.

1.8 Algorithms

Algorithms that are used or proposed should be described in pseudocode, with the same syntax and format throughout the thesis. The `algorithm2e` package provides the `algorithm` environment, with its own caption and label, and commands for conditionals, loops, etc. Listing 1.10 produces Algorithm 1.1.

Algorithm 1.1. Mini-batch gradient descent.

Input: Training set $\mathcal{D}$, learning rate η, number of epochs E
Output: Trained parameters $\boldsymbol{\theta}$
Initialise $\boldsymbol{\theta}$ randomly;
for $e \leftarrow 1$ **to** E **do**
    Shuffle $\mathcal{D}$;
    foreach *mini-batch* $\mathcal{B} \subset \mathcal{D}$ **do**
        $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \, \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}; \mathcal{B})$;
    if *the validation loss has not improved for 10 epochs* **then**
        break;
return $\boldsymbol{\theta}$;

```
\begin{algorithm}[htbp]
  \caption{Mini-batch gradient descent.}
  \label{alg:gradient-descent}
  \KwIn{Training set $\mathcal{D}$, learning rate $\eta$, number of epochs $E$}
  \KwOut{Trained parameters $\bm{\theta}$}
  Initialise $\bm{\theta}$ randomly\;
  \For{$e \leftarrow 1$ \KwTo $E$}{
    Shuffle $\mathcal{D}$\;
    \ForEach{mini-batch $\mathcal{B} \subset \mathcal{D}$}{
      $\bm{\theta} \leftarrow \bm{\theta} - \eta \, \nabla_{\bm{\theta}} \mathcal{L}
    ↪ }(\bm{\theta}; \mathcal{B})$\;
    }
    \If{the validation loss has not improved for 10 epochs}{
      \textbf{break}\;
    }
  }
  \Return{$\bm{\theta}$}\;
\end{algorithm}
```

Listing 1.10. Describing an algorithm.

Note that each instruction ends with `\;` and that the caption of an algorithm appears at the top. The documentation of `algorithm2e` (https://ctan.org/pkg/algorithm2e) describes all the available commands, although a small subset is usually enough.

1.9 Source code

Besides pseudocode, a thesis may include listings of real code. All the listings in this chapter have been produced with the `lstlisting` environment of the `listings` package, which takes three main parameters: `language`, which enables syntax highlighting; `caption`; and `label`. Listing 1.11 shows a Python example. Complete source files can be included directly from disk, without copying their content, with the `\lstinputlisting` command:

```
\lstinputlisting[language=Python, caption={...}, label=lst:...]{code/file.py}
```

```python
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_score

# 10-fold cross-validation of a random forest
model = RandomForestClassifier(n_estimators=500, random_state=2026)
scores = cross_val_score(model, X, y, cv=10, scoring="f1")
print(f"F1 = {scores.mean():.3f} ± {scores.std():.3f}")
```

Listing 1.11. Training a classifier with scikit-learn.

The list of supported languages, and how to define new ones, can be found at https://en.wikibooks.org/wiki/LaTeX/Source_Code_Listings. Include only the code that helps to understand the thesis; the complete code belongs to a repository referenced from the text (see Section 5.7).

1.10 Acronyms

Acronyms are defined once in `frontmatter/acronyms.tex`, as in Listing 1.12, and used in the text with `\gls{key}`.

```
\newacronym{gpu}{GPU}{graphics processing unit}
```

Listing 1.12. Defining an acronym.

The first time an acronym is used, its long form is written followed by the short form in brackets; afterwards only the short form appears, and every acronym used is added to the List of Acronyms automatically. For example, the first `\gls{ml}` of this chapter produces ‘machine learning (ML)’, and the next one produces ‘ML’. The plural is obtained with `\glspl{key}`, as in ‘graphics processing units (GPUs)’.

1.11 Citations and references

During the thesis you will study books, scientific articles, theses, standards and web pages, and you may use definitions, data or figures taken from them. In all these cases it is essential to cite the source properly. Poor citation practice is one of the most common reasons for a lower grade, and plagiarism, even unintentional, can lead to failing the thesis.

1.11.1 Bibliographic references

References are cited with `\cite{key}`, where `key` is the identifier of the corresponding entry in `references.bib`. The process is:

1. Get the bibliographic data in BibTeX format: from Google Scholar (the 'Cite' button, then 'BibTeX'), from the publisher's page, from `https://doi.org` (which returns BibTeX for any Digital Object Identifier (DOI)), or from a reference manager such as Zotero with the Better BibTeX extension, which keeps `references.bib` up to date automatically.
2. Add the entry to `references.bib` and give it an identifier that is easy to remember, e.g. `surname2024keyword`.
3. Cite it in the text with `\cite{identifier}`.

Each type of work has its own type of entry: `@article`, `@book`, `@inproceedings`, `@thesis`, `@online`, `@software`, `@dataset`... In general, every entry needs at least `author`, `title` and `year` (or `date`); books also need the `publisher`; articles, the `journal`, `volume`, `number` and `pages`; web resources, the `url` and the date when you accessed them (`urldate`). Always add the `doi` when the work has one. Automatically generated entries are often incomplete or contain errors, so review them carefully. Listing 1.13 shows four entries of different types, which you can also find in `references.bib`.

```
@article{charte2020review,
  title   = {A Comprehensive and Didactic Review on Multilabel
             Learning Software Tools},
  author  = {Charte, Francisco},
  journal = {IEEE Access},
  volume  = {8},
  pages   = {50330--50354},
  year    = {2020},
  doi     = {10.1109/ACCESS.2020.2979787}
}
@book{herrera2016multilabel,
  title     = {Multilabel Classification: Problem Analysis, Metrics
               and Techniques},
  author    = {Herrera, Francisco and Charte, Francisco and
```

```
                Rivera, Antonio J. and del Jesus, Mar{\'\i}a J.},
  year      = {2016},
  publisher = {Springer},
  isbn      = {978-3-319-41110-1},
  doi       = {10.1007/978-3-319-41111-8}
}
@inproceedings{charte2013imbalance,
  title     = {A First Approach to Deal with Imbalance in
               Multi-label Datasets},
  author    = {Charte, Francisco and Rivera, Antonio and
               del Jesus, Mar{\'\i}a Jos{\'e} and Herrera, Francisco},
  booktitle = {Hybrid Artificial Intelligent Systems (HAIS 2013)},
  pages     = {150--160},
  year      = {2013},
  doi       = {10.1007/978-3-642-40846-5_16}
}
@online{charte2019distribution,
  author  = {Charte, Francisco},
  title   = {Cómo analizar la distribución de los datos con {R}},
  year    = {2019},
  url     = {https://fcharte.com/tutoriales/20170114-DistribucionDatosR/},
  urldate = {2020-09-30},
  langid  = {spanish}
}
```

Listing 1.13. BibLaTeX entries of different types.

With the entries ready, the paragraph in Listing 1.14 produces the text that follows it, and the complete references are added automatically, correctly formatted, to the References section at the end of the thesis. Only cited works appear there.

```
As explained in~\cite{charte2020review} and extended in the
book~\cite{herrera2016multilabel}, the techniques presented
in~\cite{charte2013imbalance} can be put into practice by following
the tutorial in~\cite{charte2019distribution}.
```

Listing 1.14. A paragraph citing several works.

As explained in [2] and extended in the book [3], the techniques presented in [4] can be put into practice by following the tutorial in [5].

The citation style is chosen with the `bibstyle` class option (Table 1.2). Numeric styles are common in engineering and computer science, while author–year styles such as APA are common in health and social sciences. Ask your supervisor which one is expected in your field. With author–year styles, use `\textcite{key}` to cite the authors as part of the sentence, e.g. 'Pedregosa et al. [6] showed that. . . ', and `\parencite{key}` for citations in parentheses.

1.11.2 Quotations and third-party material

If, besides using a work as a source, you reproduce part of it literally, follow these rules:

- Only short fragments of a work can be quoted literally, usually no more than a paragraph and never whole pages.
- Short quotations go in the text between quotation marks, with `\enquote{...}`; longer quotations are set apart with the `displayquote` environment:

  > This is how a long literal quotation looks: it is indented and separated from the rest of the text, so that the reader can clearly tell the words of the author apart from yours. [2]

- Every quotation must be accompanied by its bibliographic citation, so that its origin can be identified unambiguously.
- Do not include illustrations, diagrams or figures whose rights of use are not clear. Material under a Creative Commons licence can be included by indicating its source and licence; otherwise you need the owner's permission. The alternative is to draw your own figure, indicating that it is based on an existing source ('adapted from [3]').

Remember that your thesis will probably be published in your university's institutional repository, where anyone can read it, and that it will be checked with plagiarism detection tools. Be honest, and never use the work of others without citing it.

1.12 Special characters

Some characters have a special meaning in LaTeX: `\` starts commands, `$` delimits mathematics, `%` starts a comment, `&` separates table columns, `#` denotes parameters, `_` and `^` create subscripts and superscripts, and `{ }` group content. To print them, precede them with a backslash: `\%`, `\$`, `\&`, `\#`, `\_`, `\{` and `\}` produce %, $, &, #, _, { and }. The backslash itself is written with `\textbackslash` (\), because `\\` means a line break. Accented letters such as á, é, ñ or ç can be typed directly. If you have trouble with any character, see `https://en.wikibooks.org/wiki/LaTeX/Special_Characters`.

1.13 Working with your supervisors

When you send a draft to your supervisors, compile it with the `review` option: every line is numbered, so comments such as 'page 12, line 34' are unambiguous, and the watermark prevents

a draft from being mistaken for the final version. While writing, you can leave notes for yourself or your supervisors with `\todo{text}`, which appears in the margin, or with `\todo[inline]{text}`:

This is a to-do note.

This is an inline to-do note. All notes disappear with the `final` option.

Use a version control system such as Git from the very first day (Overleaf can synchronise with GitHub), so that you never lose work and can always go back to a previous version.

1.14 Structure of a Master's thesis

Once your data are set, the next step is writing the content. The structure depends on the type of thesis and on the regulations of your programme, so take the following as a recommendation that can be adapted to each case. Each chapter is stored in its own `.tex` file and included from `main.tex` with `\input`: add, remove or rename the files as you need.

1.14.1 Research thesis

This is the most common type of Master's thesis, and the one this template is prepared for. Its chapters follow the usual structure of a scientific article:

1. **Introduction** — `01-introduction.tex`. A high-level overview of the work: motivation, importance of the problem, how it has been addressed so far and what this thesis contributes, without going into detail. It usually ends with the structure of the document.
2. **Background and Related Work** — `02-background.tex`. Shows that you have studied the topic: the problem in detail, the techniques used so far and those you propose. The bibliography is fundamental in this chapter.
3. **Objectives** — `03-objectives.tex`. The general objective, broken down into specific ones, and, if appropriate, the hypothesis or research questions. Just by reading this chapter, it should be clear what has been done.
4. **Materials and Methods** — `04-materials-and-methods.tex`. The data and resources used and the techniques applied, in enough detail to reproduce the work, including the experimental setup, the evaluation and, when relevant, the ethical and legal aspects.
5. **Results** — `05-results.tex`. All the results obtained, usually in tables and figures, compared with other methods whenever possible.
6. **Discussion** — `06-discussion.tex`. The interpretation of the results, their comparison with previous work, their limitations and their implications. Some fields merge it with the previous chapter.

7. **Conclusions and Future Work** — `07-conclusions.tex`. A summary of the work and results, an assessment of whether the objectives of the approved proposal have been achieved, and the lines of future work.

1.14.2 Professional or engineering project

In professionally oriented Master's degrees, the thesis is often a project: the development of a software system, a product or a solution for an organisation. Its usual chapters are:

1. **Introduction**, as described above.
2. **Planning and Budget**: time and cost planning of the project and the resources it needs; the methodology used to manage it (e.g. Scrum); the estimation of the size and effort of the software, from which the schedule (usually a Gantt chart, see Figure 5.1) and the budget are derived; and the legal and technical regulations followed.
3. **Requirements Analysis**: the interviews with the (real or hypothetical) users or clients, from which the functional and non-functional requirements are identified, analysed, specified (use case diagrams, user stories, domain model. . .) and validated, following the practices recommended in the ISO/IEC/IEEE 29148 standard.
4. **System Design**: the detailed design of the system core (architecture, package, class and sequence diagrams), its user interface (wireframes or mockups) and its data (entity–relationship diagrams or equivalent), including the architectural and design patterns used.
5. **Implementation**: the evaluation of the alternative languages, libraries and tools and the justified choice among them; the implementation iterations; the structure of the project; the coding and documentation style guides; and the build and configuration process.
6. **Testing**: the tools used and the unit, integration, regression and acceptance tests carried out.
7. **Deployment and Maintenance**: how the system is deployed and put into operation, its dependencies on external hosting or services, and the operations needed to maintain it.
8. **Conclusions and Future Work**, as described above.
9. **Installation and User Manuals**, usually as appendices, written for the final users rather than for developers.

Projects of this type need diagrams of several kinds. Tools such as PlantUML, which define diagrams as text files that can be kept under version control, are highly recommended.

1.15 Writing advice

Although it has nothing to do with LaTeX, the quality of the writing deserves a final word. For the examination board and for any future reader, the quality of the text is a reflection of the quality of the work. A very good piece of work can be penalised because of poor writing, while a less brilliant one with an impeccable presentation may be valued more highly. Without trying to be exhaustive, keep in mind the following:

- **Spelling:** choose British or American English (option `english`) and be consistent: *analyse*/*analyze*, *modelling*/*modeling*, *colour*/*color*. Use a spell checker, such as the one integrated in Overleaf, set to the same variety.
- **Grammar and tense:** use the present tense for established knowledge and for what the thesis contains ('Chapter 6 presents…') and the past tense for what you did ('the data were collected…'). Avoid contractions (*don't*, *it's*) in formal writing.
- **Common pitfalls for Spanish speakers:** watch out for false friends: *actually* means 'en realidad' (not 'actualmente', which is *currently*), *eventually* means 'finalmente', and *sensible* means 'sensato'. Do not use 'the' with general plural nouns ('*Patients* with diabetes…', not '*The* patients…'), adjectives never take a plural *-s*, and experiments are *conducted* or *carried out*, not *made*. Acronyms do take an *s* in the plural in English (*GPUs*).
- **Vocabulary:** use precise, specialised terms, which show that you know the field, and vary your vocabulary to avoid repeating the same words. A thesis is a formal document: write 'as described in…' rather than 'as we said before…'.
- **Structure:** the thesis as a whole, each chapter and each section must read as a sequence of connected ideas. Nothing is more confusing than a text in which each paragraph jumps to a different topic. Plan the content of each chapter and section before writing it so that, as in good software design, cohesion is maximised and coupling minimised.
- **Generative AI:** if you use AI tools to improve your writing, follow the rules of your university and declare their use (see `frontmatter/ai-statement.tex`). You remain the author and the person responsible for every sentence of the thesis.

One last piece of advice, completely outside the scope of these instructions: the best way to learn to write well is to read a lot, especially good scientific articles in your field and good literature.

1.16 Getting this template

This template is publicly available at https://github.com/thisisjosepablo/TFMLatexTemplate, where you can download the latest release, open it directly in Overleaf, report problems or contribute improvements, such as settings that reproduce the style of your own university. It is released into the public domain under the CC0 1.0 licence, so you can use it however you like, without any condition, but also without any warranty.

CHAPTER 2

Introduction

Template guidance

The introduction is a high-level overview of the whole work. Explain the motivation and why the problem matters, how it has been approached so far (briefly; the details belong to Chapter 3), what is missing and what this thesis contributes, without going into technical detail. Finish with the structure of the document, one or two sentences per chapter. Write it, or at least revise it, at the end, when you know exactly what the thesis contains.

2.1 Motivation

[Describe the context of the problem and why it is relevant: scientific, social, clinical, economic or industrial impact. Support your claims with references.]

2.2 Problem statement

[State clearly and concisely the problem addressed in this thesis and the gap in the current knowledge or practice that it aims to fill.]

2.3 Contributions

[List the main contributions of the thesis. Be concrete: a new method, a dataset, an empirical comparison, a software tool, a validated prototype. . .]

2.4 Structure of the document

The rest of this document is organised as follows. Chapter 3 reviews the theoretical background and the state of the art. Chapter 4 defines the objectives of the thesis. Chapter 5 describes the materials and methods, and Chapter 6 presents the results, which are discussed in Chapter 7. Finally, Chapter 8 summarises the conclusions and outlines future work.

CHAPTER 3

Background and Related Work

Template guidance

Show that you know the field. First describe the problem and the concepts and techniques needed to understand the thesis (theoretical framework); then review critically the most relevant work published on the topic (state of the art), grouping it by approach rather than listing papers one by one. End by identifying the gap that your work addresses. References are essential here: search the scientific literature (Scopus, Web of Science, Google Scholar, PubMed, IEEE Xplore, ACM Digital Library…) and cite every source.

3.1 Theoretical framework

[Introduce the concepts, definitions and techniques on which the thesis is built. For example, an introduction to ML for multi-label problems can be found in [3].]

3.2 State of the art

[Review and compare the most relevant previous works. A summary table that compares the approaches along a few dimensions is often very useful.]

3.3 Summary and research gap

[Summarise what is known and, most importantly, what is not known yet or not solved, which motivates the objectives in Chapter 4.]

CHAPTER 4

Objectives

Template guidance

State the general objective of the thesis from a high-level perspective, and break it down into specific, measurable objectives. A reader should get a clear idea of what has been done just by reading this chapter. If your work is hypothesis-driven, state the hypothesis or research questions first. The specific objectives will be revisited in Chapter 8 to assess whether they have been achieved.

4.1 Research questions

RQ1. [First research question]

RQ2. [Second research question]

4.2 General objective

[One sentence describing the overall aim of the thesis.]

4.3 Specific objectives

O1. [Review the state of the art on . . .]

O2. [Collect and prepare the data . . .]

O3. [Design and implement . . .]

O4. [Evaluate . . . and compare with . . .]

4.4 Scope and limitations

[Define what is within the scope of the thesis and what is deliberately left out.]

CHAPTER 5

Materials and Methods

Template guidance

Describe, in enough detail for another person to reproduce your work, the *materials* (data, participants, equipment, software) and the *methods* (techniques, algorithms, experimental design, evaluation) used to achieve the objectives. Justify every choice. Explain how the data were obtained and prepared, which techniques were used and why, how the experiments were run (hardware and software configuration) and how the results were evaluated. If your work involves people or personal data, the ethical and legal section is mandatory.

5.1 Work plan

[Describe the methodology followed to organise the work and its time plan. Figure 5.1 shows an example of a Gantt chart drawn with the `pgfgantt` package.]

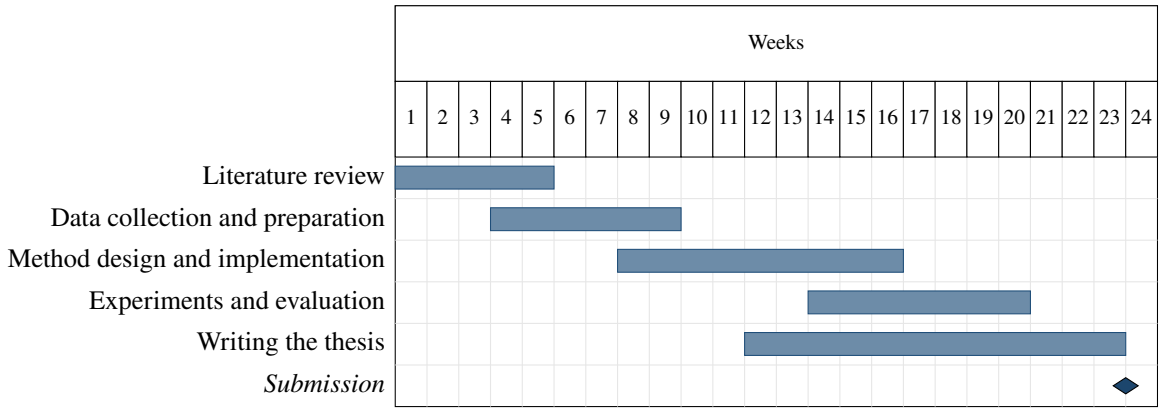


Figure 5.1. Example time plan of the thesis.

5.2 Data

[Source of the data, description (size, variables, classes, period), quality issues and pre-processing. A table describing the variables is usually helpful.]

5.3 Methods

[Techniques and algorithms used, with references and the justification of the choice.]

5.4 Experimental setup

[Hardware and software configuration, hyperparameters, random seeds, validation strategy. Table 5.1 is an example of a hardware requirements table.]

Table 5.1. Hardware requirements of the computer on which the system will be installed.

Component	Specification	Description
Processor	x86-64 architecture	Latest-generation multi-core processor, able to run the containers that make up the system in parallel
RAM	64 GB	Enough memory is essential for the system to work correctly
Storage	10 TB	Mass storage for the images, videos, AI models and other elements of the system; at least 7 200 rpm is recommended
GPU	CUDA CC $\geq$ 3.5	GPU with CUDA Compute Capability 3.5 or later and at least 8 GB of memory to run the AI models
Network	Gigabit Ethernet	Access to external data sources and content sharing

5.5 Evaluation

[Metrics, baselines, statistical tests and how the results are compared.]

5.6 Ethical and legal considerations

[If the work uses personal data, describe how the General Data Protection Regulation (GDPR) [7] and the Spanish Organic Law 3/2018 on the Protection of Personal Data and Guarantee of Digital Rights [8] have been complied with: anonymisation or pseudonymisation, legal basis, informed consent, storage and access. Mention the approval of the ethics committee, if needed, with its reference number.]

5.7 Reproducibility

[State where the code and, if possible, the data can be found (a public repository with a DOI, e.g. on Zenodo), the software versions used [6] and any restriction on data access.]

CHAPTER 6

Results

Template guidance

Report all the results obtained by applying the methods to the materials, usually in tables and figures, in the same order as the experiments were described in Chapter 5. Describe what the results show objectively, without interpreting them yet: the interpretation belongs to Chapter 7 (some theses merge both chapters into a single 'Results and Discussion' chapter). Whenever possible, compare different methods and include measures of dispersion and statistical significance.

6.1 [First experiment]

[Results of the first experiment. Table 6.1 shows an example of a results table with numbers aligned on the decimal point using `siunitx`; the best result of each column is highlighted in bold.]

Table 6.1. Example of a results table: mean ± standard deviation over 10 runs. The best value of each column is shown in bold.

Method	Accuracy	F1 score	Time (s)
Baseline	0.812 ± 0.014	0.774 ± 0.019	12.4
Method A	0.845 ± 0.011	0.801 ± 0.016	48.9
Method B	0.851 ± 0.009	0.815 ± 0.012	103.2
Proposed method	$\mathbf{0.873 \pm 0.008}$	$\mathbf{0.842 \pm 0.010}$	61.7

6.2 [Second experiment]

[Results of the second experiment.]

6.3 Comparison with the state of the art

[Compare your results with those published in the literature reviewed in Chapter 3.]

CHAPTER 7

Discussion

Template guidance

Interpret the results: what do they mean, how do they answer the research questions, how do they compare with previous work and why. Be honest about the limitations of the study and about any threats to the validity of the conclusions. Discuss the practical implications of the findings.

7.1 Interpretation of the results

[Explain the results of Chapter 6 and answer RQ1 and RQ2.]

7.2 Comparison with previous work

[Relate your findings to the literature.]

7.3 Limitations

[Limitations of the data, the methods and the evaluation.]

7.4 Implications

[Practical, scientific, social or clinical implications of the work.]

CHAPTER 8

Conclusions and Future Work

Template guidance

Summarise the work done and the main results, and state clearly whether the objectives set in Chapter 4 (those of the thesis proposal approved by the academic committee) have been achieved. Do not introduce new results here. Finish with the lines of future work that derive from the thesis. Some universities also ask for a personal assessment of the competences acquired.

8.1 Conclusions

[Main conclusions of the thesis.]

8.2 Fulfilment of the objectives

Table 8.1 relates each specific objective to the part of the thesis where it has been addressed.

Table 8.1. Fulfilment of the specific objectives of the thesis.

Objective	How it has been achieved	Where
O1	[Brief description]	Chapter 3
O2	[Brief description]	Section 5.2
O3	[Brief description]	Section 5.3
O4	[Brief description]	Chapter 6

8.3 Future work

[Possible improvements and new lines of research that derive from this work.]

References

[1] Ministerio de Universidades. *Real Decreto 822/2021, de 28 de septiembre, por el que se establece la organización de las enseñanzas universitarias y del procedimiento de aseguramiento de su calidad*. Boletín Oficial del Estado, no. 233. 29th Sept. 2021. URL: https://www.boe.es/eli/es/rd/2021/09/28/822.

[2] F. Charte. 'A Comprehensive and Didactic Review on Multilabel Learning Software Tools'. In: *IEEE Access* 8 (2020), pp. 50330–50354. DOI: 10.1109/ACCESS.2020.2979787.

[3] F. Herrera, F. Charte, A. J. Rivera and M. J. del Jesus. *Multilabel Classification: Problem Analysis, Metrics and Techniques*. Cham: Springer, 2016. DOI: 10.1007/978-3-319-41111-8.

[4] F. Charte, A. Rivera, M. J. del Jesus and F. Herrera. 'A First Approach to Deal with Imbalance in Multi-label Datasets'. In: *Hybrid Artificial Intelligent Systems (HAIS 2013)*. Vol. 8073. Lecture Notes in Computer Science. Springer, 2013, pp. 150–160. DOI: 10.1007/978-3-642-40846-5_16.

[5] F. Charte. *Cómo analizar la distribución de los datos con R*. 2019. URL: https://fcharte.com/tutoriales/20170114-DistribucionDatosR/ (visited on 30/09/2020).

[6] F. Pedregosa et al. 'Scikit-learn: Machine Learning in Python'. In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.

[7] European Parliament and Council of the European Union. *Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation)*. Official Journal of the European Union, L 119. 4th May 2016. URL: https://eur-lex.europa.eu/eli/reg/2016/679/oj.

[8] Jefatura del Estado. *Ley Orgánica 3/2018, de 5 de diciembre, de Protección de Datos Personales y garantía de los derechos digitales*. Boletín Oficial del Estado, no. 294. 6th Dec. 2018. URL: https://www.boe.es/eli/es/lo/2018/12/05/3.

APPENDIX A

Alignment with the Sustainable Development Goals

Template guidance

Several Spanish universities ask students to reflect on how their thesis relates to the SDGs of the United Nations 2030 Agenda. Mark the degree of relationship of your work with each goal by changing the last argument of each `\sdg` command to `H` (high), `M` (medium), `L` (low) or `N` (not applicable), and then justify the most relevant ones. If your university does not ask for it, remove this appendix from `main.tex`.

Table A.1. Degree of relationship of this thesis with the Sustainable Development Goals.

Sustainable Development Goal	High	Medium	Low	Not applicable
SDG 1. No poverty				✓
SDG 2. Zero hunger				✓
SDG 3. Good health and well-being	✓			
SDG 4. Quality education			✓	
SDG 5. Gender equality				✓
SDG 6. Clean water and sanitation				✓
SDG 7. Affordable and clean energy				✓
SDG 8. Decent work and economic growth				✓
SDG 9. Industry, innovation and infrastructure		✓		
SDG 10. Reduced inequalities			✓	
SDG 11. Sustainable cities and communities				✓
SDG 12. Responsible consumption and production				✓
SDG 13. Climate action				✓
SDG 14. Life below water				✓
SDG 15. Life on land				✓
SDG 16. Peace, justice and strong institutions				✓
SDG 17. Partnerships for the goals				✓

[Explain how the thesis contributes to the goals marked as high or medium, and its possible negative impacts, if any.]

APPENDIX B

Supplementary Material

Template guidance

Appendices contain material that supports the thesis but would interrupt the reading of the main text: additional tables and figures, questionnaires, consent forms, detailed proofs, user manuals, extended code listings, etc. Appendices are numbered with letters and referenced from the text like chapters, e.g. Appendix A. Whole PDF documents can be inserted with `\includepdf[pages=-]{file.pdf}` from the `pdfpages` package.

B.1 Source code

[Link to the repository where the code of this thesis is available, e.g. `https://github.com/username/repository`, and to its archived version with a DOI.]

B.2 Additional results

[Additional tables and figures.]